# Condensed Matter Physics

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## Lecture Notes: Energy Bands in Solids

(based on Mod-01 Lec-34, "Energy Bands in Solids")

**Note on these notes.** These notes reconstruct the standard derivation sequence for "Energy Bands in Solids" as covered in this course (free-electron → Bloch electron → Kronig–Penney model → nearly-free-electron model → tight-binding model), with every algebraic step shown, including steps that are often left as an exercise on the blackboard. Wherever the lecture uses a specific hand-drawn figure or numerical sketch that cannot be reconstructed exactly, an empty boxed placeholder is left for you to fill in from your own viewing of the video. Two graphs (the Kronig–Penney transcendental function and the free-electron/reduced-zone dispersion) are produced numerically with Python since a computed curve is more accurate here than a hand-drawn TikZ sketch.

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# 1 From Atomic Levels to Energy Bands

## 1.1 Limitations of the free-electron theory

The free-electron (Sommerfeld) theory is a genuine success in several respects — it accounts reasonably well for the electronic specific heat, thermal conductivity, and electrical conductivity of simple metals. Its starting assumptions, however, are severe:

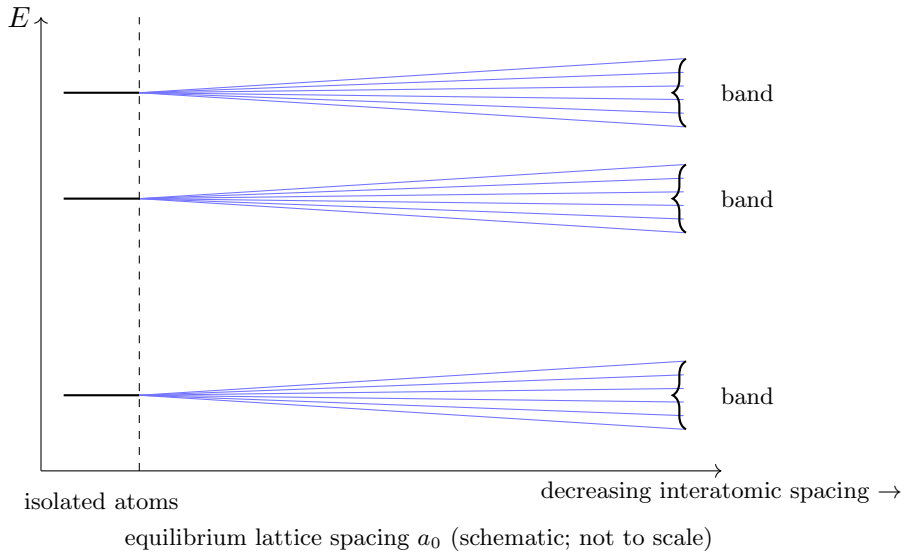
- It invokes the **one-electron approximation**: each electron is treated as moving independently in an *average* potential, with electron–electron and electron–ion interactions replaced by a featureless flat potential well that merely confines electrons to the metal (infinite walls at the sample boundary, zero potential inside).
- Because the potential inside is taken to be perfectly flat, the theory cannot distinguish a **metal from a semiconductor from an insulator** — all solids would conduct equally well.
- It cannot account for the observed **positive Hall coefficient** of several metals, nor for the details of magnetotransport (cyclotron resonance, de Haas–van Alphen oscillations, etc.).

The fix is to keep the one-electron approximation (each electron still moves independently) but replace the flat well by the *true, periodic* potential of the ion lattice. That periodic potential is the source of all the band structure derived in this lecture.

## 1.2 Motivation

In an isolated atom the electrons occupy sharp, discrete energy levels $E_1, E_2, E_3, \ldots$, each of which can hold a fixed number of electrons (2 per orbital, by the Pauli principle, including spin). When $N$ identical atoms are brought together to form a crystal:

- The outer (valence) electron wavefunctions of neighbouring atoms begin to overlap.
- By the Pauli exclusion principle, no two electrons in the *combined* system can have identical quantum numbers, so each $N$-fold degenerate atomic level must split into $N$ distinct, closely-spaced levels.
- For $N \sim 10^{23}$ (a macroscopic crystal) and a level spread of order a few eV, the spacing between adjacent levels is of order $10^{-23}$ eV — so close that the set of levels is, for all practical purposes, a **continuous energy band**.
- Adjacent bands (arising from different atomic shells) may be separated by a **forbidden gap** $E_g$, a range of energy for which no traveling-wave solution exists in the crystal.




*[Blank space: insert the lecturer's specific board sketch here – e.g. the $E$ vs. $1/r$ curve for Na $3s, 3p$ levels broadening into bands as drawn in the lecture, if it differs from the schematic above.]*

### 1.3 The central problem

To go beyond this qualitative picture we must solve the Schrödinger equation for an electron moving in the *periodic potential* $V(\mathbf{r})$ created by the regular array of ion cores,

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r})\right]\psi(\mathbf{r}) = E\,\psi(\mathbf{r}), \qquad V(\mathbf{r}) = V(\mathbf{r}+\mathbf{R}_n) \tag{1}$$

where $\mathbf{R}_n$ is any real-space lattice translation vector,

$$\mathbf{R}_n = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3, \qquad n_1, n_2, n_3 \in \mathbb{Z}, \tag{2}$$

built from the primitive translation vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ of the lattice. Three levels of approximation are used in this lecture to solve (1):

1. An *exactly solvable* periodic potential — the Kronig–Penney square-well model (Section 3).
2. Perturbation theory starting from *free electrons*, valid when $V(\mathbf{r})$ is weak — the nearly-free-electron model (Section 4).
3. Perturbation theory starting from *isolated-atom orbitals*, valid when the overlap between neighbouring atoms is weak — the tight-binding (LCAO) model (Section 5).

All three must, and do, lead to the same qualitative conclusion: allowed energy bands separated by forbidden gaps.

### 1.4 Qualitative origin of bands: the diatomic-molecule analogy

Before doing the full periodic-potential problem it is worth seeing, in the simplest possible model, *why* bringing atoms together must split their levels. Consider two identical isolated atoms $A$ and $B$, infinitely far apart, each with a single relevant electron in a level of energy $E_0$, so that

$$H_{AA} \equiv \langle A|H|A\rangle = E_0, \qquad H_{BB} \equiv \langle B|H|B\rangle = E_0.$$

The two atomic levels are degenerate: $E_A = E_B = E_0$.

Now bring the atoms together so that their wavefunctions overlap. An interatomic (exchange/hopping) matrix element switches on,

$$H_{AB} \equiv \langle A|H|B\rangle = H_{BA}^*,$$

and the true eigenstates of the two-level system are found from the secular determinant

$$\begin{vmatrix} H_{AA} - E & H_{AB} \\ H_{BA} & H_{BB} - E \end{vmatrix} = 0. \tag{3}$$

Expanding,

$$(H_{AA} - E)(H_{BB} - E) - H_{AB}H_{BA} = 0 \implies E^2 - (H_{AA} + H_{BB})E + H_{AA}H_{BB} - |H_{AB}|^2 = 0,$$

with roots

$$E = \frac{H_{AA} + H_{BB}}{2} \pm \frac{1}{2}\sqrt{(H_{AA} - H_{BB})^2 + 4|H_{AB}|^2}. \tag{4}$$

For identical atoms, $H_{AA} = H_{BB} = E_0$, and (4) collapses to

$$\boxed{E_1 = E_0 + |H_{AB}|, \qquad E_2 = E_0 - |H_{AB}|.} \tag{5}$$

The single degenerate atomic level $E_0$ has split into two distinct levels, separated by $2|H_{AB}|$ — exactly the same structure that will reappear as the nearly-free-electron gap in Section 4 and the bandwidth $4\beta_{TB}$ in the tight-binding model of Section 5. (This is, in fact, the $N = 2$ special case of the tight-binding secular problem worked out in general in Section 5.)

**Generalizing to $N \sim 10^{23}$ atoms.** With $N$ atoms brought together into a regular lattice, the $N \times N$ secular determinant analogous to (3) has $N$ roots: the single $N$-fold-degenerate atomic level splits into $N$ distinct, non-degenerate sub-levels. For a macroscopic crystal ($N \sim 10^{23}$) confined within a level spread of a few eV, adjacent sub-levels are separated by only $\sim 10^{-23}$ eV — a spacing utterly unresolvable in any measurement — so the set of $N$ discrete levels behaves, for all practical purposes, as a continuous **energy band**. This is the qualitative picture; the rest of this lecture derives it quantitatively.

## 2 Bloch's Theorem

**Statement.** For an electron moving in a strictly periodic potential $V(x+a) = V(x)$ (1-D, lattice constant $a$), the eigenstates of (1) can always be chosen to have the form

$$\psi_k(x) = e^{ikx}\, u_k(x), \qquad u_k(x + a) = u_k(x) \tag{6}$$

i.e. a plane wave modulated by a function with the periodicity of the lattice. Equivalently,

$$\psi_k(x + a) = e^{ika}\, \psi_k(x). \tag{7}$$

### 2.1 Derivation via Fourier expansion (the "central equation")

This is the route actually used in the lecture, and it is worth doing in full because the intermediate object — the *central equation* — is reused conceptually in the nearly-free-electron model of Section 4.

**Step 1: Fourier-expand the periodic potential.** Since $V(\mathbf{r}) = V(\mathbf{r} + \mathbf{R}_n)$, it has a Fourier series over the reciprocal lattice vectors $\mathbf{G}$ (defined by $\mathbf{G} \cdot \mathbf{R}_n = 2\pi \times$ integer, so that $e^{i\mathbf{G}\cdot\mathbf{R}_n} = 1$):

$$V(\mathbf{r}) = \sum_{\mathbf{G}} V_{\mathbf{G}}\, e^{i\mathbf{G}\cdot\mathbf{r}}. \tag{8}$$

**Step 2: Expand the unknown wavefunction in plane waves.** Since plane waves $e^{i\mathbf{k}\cdot\mathbf{r}}$ form a complete basis for any function on the crystal (with periodic boundary conditions),

$$\psi(\mathbf{r}) = \sum_{\mathbf{k}} C(\mathbf{k})\, e^{i\mathbf{k}\cdot\mathbf{r}}. \tag{9}$$

**Step 3: Substitute both into the Schrödinger equation.** Acting with $-\frac{\hbar^2}{2m}\nabla^2$ on (9) brings down a factor $\hbar^2k^2/2m$ from each term; acting with $V(\mathbf{r})$ multiplies term by term:

$$\sum_{\mathbf{k}} \frac{\hbar^2k^2}{2m} C(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}} + \sum_{\mathbf{k}}\sum_{\mathbf{G}} V_{\mathbf{G}}\, C(\mathbf{k})\, e^{i(\mathbf{k}+\mathbf{G})\cdot\mathbf{r}} = E \sum_{\mathbf{k}} C(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}}. \tag{10}$$

**Step 4: Collect terms under a common phase.** In the second (double) sum, relabel the dummy wavevector $\mathbf{k} \to \mathbf{k} - \mathbf{G}$ so that every term in (10) carries the same phase factor $e^{i\mathbf{k}\cdot\mathbf{r}}$:

$$\sum_{\mathbf{k}} \left\{ \frac{\hbar^2k^2}{2m} C(\mathbf{k}) + \sum_{\mathbf{G}} V_{\mathbf{G}}\, C(\mathbf{k}-\mathbf{G}) \right\} e^{i\mathbf{k}\cdot\mathbf{r}} = \sum_{\mathbf{k}} E\, C(\mathbf{k})\, e^{i\mathbf{k}\cdot\mathbf{r}}.$$

Because the plane waves $e^{i\mathbf{k}\cdot\mathbf{r}}$ are linearly independent, the coefficient of *each* $\mathbf{k}$ must vanish separately. This gives the **central equation**:

$$\boxed{\left(\frac{\hbar^2k^2}{2m} - E\right) C(\mathbf{k}) + \sum_{\mathbf{G}} V_{\mathbf{G}}\, C(\mathbf{k}-\mathbf{G}) = 0.} \tag{11}$$

This is nothing but the Schrödinger equation rewritten in $\mathbf{k}$-space: it says the amplitude $C(\mathbf{k})$ is coupled *only* to amplitudes $C(\mathbf{k}-\mathbf{G})$ at wavevectors differing from $\mathbf{k}$ by a reciprocal lattice vector — never to any other $\mathbf{k}'$.

**Step 5: Read off Bloch's theorem.** Equation (11) shows that, for a fixed "label" $\mathbf{k}$, the full eigenstate built from it involves only the discrete, coupled set of plane waves $\{\mathbf{k}-\mathbf{G}\}_{\mathbf{G}}$:

$$\psi_{\mathbf{k}}(\mathbf{r}) = \sum_{\mathbf{G}} C(\mathbf{k}-\mathbf{G})\, e^{i(\mathbf{k}-\mathbf{G})\cdot\mathbf{r}} = \left[\sum_{\mathbf{G}} C(\mathbf{k}-\mathbf{G})\, e^{-i\mathbf{G}\cdot\mathbf{r}}\right] e^{i\mathbf{k}\cdot\mathbf{r}}. \tag{12}$$

Define the bracketed quantity as $u_{\mathbf{k}}(\mathbf{r})$. Since $e^{-i\mathbf{G}\cdot\mathbf{R}_n} = 1$ for every reciprocal lattice vector $\mathbf{G}$ and every real-space lattice vector $\mathbf{R}_n$,

$$u_{\mathbf{k}}(\mathbf{r}+\mathbf{R}_n) = \sum_{\mathbf{G}} C(\mathbf{k}-\mathbf{G})\, e^{-i\mathbf{G}\cdot(\mathbf{r}+\mathbf{R}_n)} = \sum_{\mathbf{G}} C(\mathbf{k}-\mathbf{G})\, e^{-i\mathbf{G}\cdot\mathbf{r}} = u_{\mathbf{k}}(\mathbf{r}),$$

i.e. $u_{\mathbf{k}}$ has exactly the periodicity of the lattice. Equation (12) is then precisely $\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{\mathbf{k}}(\mathbf{r})$ — **Bloch's theorem**, Eq. (6), derived directly from the Schrödinger equation with no further assumptions. ■

### 2.2 Alternative (operator) proof

It is also useful to know the shorter, more abstract group-theoretic proof. Define the translation operator $T_a$ by $T_a f(x) = f(x+a)$ (1-D, for brevity). Because $V(x)$ is periodic, $T_a$ commutes with the Hamiltonian $H = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x)$:

$$[H, T_a] = 0.$$

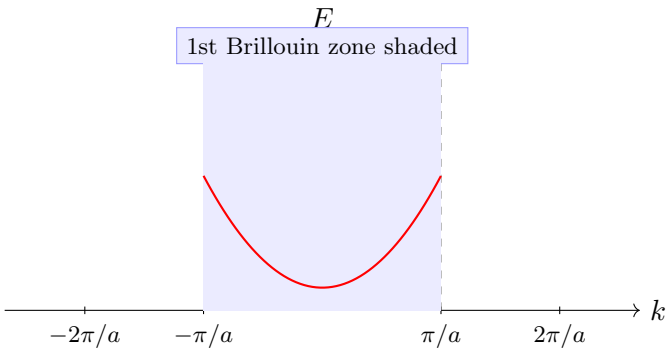
Hence $H$ and $T_a$ possess a common set of eigenfunctions. Since translations form an abelian group, the eigenvalues of the unitary operator $T_a$ must be pure phases; write them as $e^{ika}$ for real $k$:

$$T_a\psi(x) = \psi(x+a) = e^{ika}\psi(x).$$

Writing $\psi(x) = e^{ikx}u(x)$ and substituting into the last relation gives $u(x+a) = u(x)$, which is (6). ■

### 2.3 Consequences used repeatedly below

- $k$ (the *crystal momentum* $\hbar k$) is a good quantum number; it labels the eigenstates.
- $k$ and $k + 2\pi/a$ describe the *same* state (since $e^{i(k+2\pi/a)a} = e^{ika}$), so it is sufficient to consider $k \in (-\pi/a, \pi/a]$, the **first Brillouin zone**.
- For each $k$ there is, in general, a discrete ladder of energies $E_n(k)$, $n = 1, 2, 3, \ldots$ — these are the **energy bands**.

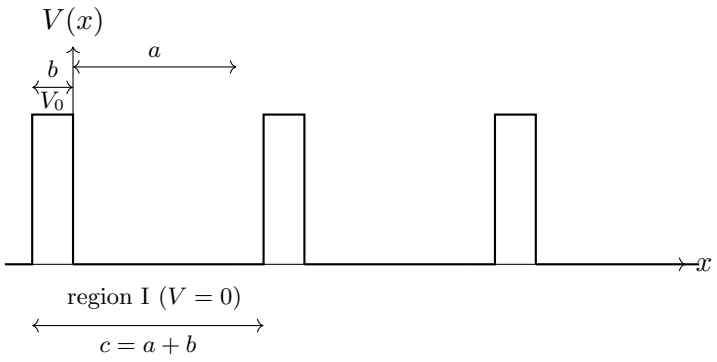



## 3 The Kronig–Penney Model

### 3.1 The model potential

Kronig and Penney (1931) replaced the real periodic potential of the ion cores by an idealised 1-D array of rectangular potential wells of width $a$ (where the electron is essentially free, $V = 0$) separated by rectangular barriers of width $b$ and height $V_0$, with period $c = a + b$:

$$V(x) = \begin{cases} 0, & 0 < x < a \quad \text{(region I)} \\ V_0, & -b < x < 0 \quad \text{(region II)} \end{cases}, \qquad V(x+c) = V(x).$$




### 3.2 Solving the Schrödinger equation in each region

**Region I** ($0 < x < a$, $V = 0$):

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} = E\psi \implies \psi_{\mathrm{I}}(x) = Ae^{i\alpha x} + Be^{-i\alpha x}, \qquad \alpha \equiv \sqrt{\frac{2mE}{\hbar^2}}. \tag{13}$$

**Region II** ($-b < x < 0$, $V = V_0 > E$):

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} + V_0\psi = E\psi \implies \psi_{\mathrm{II}}(x) = Ce^{\beta x} + De^{-\beta x}, \qquad \beta \equiv \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}. \tag{14}$$

(If instead $E > V_0$, $\beta \to i\alpha'$ and the solutions become oscillatory; the algebra below goes through with $\cosh \to \cos$, $\sinh \to \sin$.)

### 3.3 The lecture's route: substitution via the periodic envelope $u_k(x)$

The derivation above (direct plane-wave solutions $\psi_{\mathrm{I}}, \psi_{\mathrm{II}}$ matched with an explicit Bloch phase $e^{ikc}$ at the far boundary) is the most transparent route. The lecture instead builds Bloch's theorem in from the start by writing $\psi(x) = e^{ikx}u_k(x)$ (Eq. (6)) and substituting *this* form directly into the Schrödinger equation in each region, solving for $u_k(x)$ rather than $\psi(x)$. The two routes are algebraically equivalent; this one is worth seeing because it is the version most directly tied to Bloch's theorem and is quicker to set up.

Write $\psi(x) = e^{ikx}u(x)$ (dropping the $k$ subscript on $u$ for brevity) and substitute into $-\frac{\hbar^2}{2m}\psi'' + V\psi = E\psi$. Using $\psi' = e^{ikx}(u' + iku)$ and $\psi'' = e^{ikx}(u'' + 2iku' - k^2u)$, the Schrödinger equation becomes an ODE for $u(x)$ alone:

$$u'' + 2ik\,u' - k^2u = -\frac{2m}{\hbar^2}\big(E - V\big)u. \tag{15}$$

**Region 1** ($0 < x < a$, $V = 0$, using $\alpha^2 = 2mE/\hbar^2$ as before):

$$\frac{\mathrm{d}^2u_1}{\mathrm{d}x^2} + 2ik\frac{\mathrm{d}u_1}{\mathrm{d}x} + (\alpha^2 - k^2)u_1 = 0. \tag{16}$$

This is a linear ODE with constant coefficients; its characteristic equation $\lambda^2 + 2ik\lambda + (\alpha^2 - k^2) = 0$ has roots $\lambda = -ik \pm i\alpha = i(\pm\alpha - k)$, giving the general solution

$$u_1(x) = A\,e^{i(\alpha - k)x} + B\,e^{-i(\alpha + k)x}. \tag{17}$$

(Check: $e^{ikx}u_1(x) = Ae^{i\alpha x} + Be^{-i\alpha x}$, recovering exactly $\psi_{\mathrm{I}}(x)$ of Eq. (13), as it must.)

**Region 2** ($-b < x < 0$, $V = V_0$, using $\beta^2 = 2m(V_0 - E)/\hbar^2$):

$$\frac{\mathrm{d}^2u_2}{\mathrm{d}x^2} + 2ik\frac{\mathrm{d}u_2}{\mathrm{d}x} - (\beta^2 + k^2)u_2 = 0, \tag{18}$$

with characteristic roots $\lambda = -ik \pm \beta$, giving

$$u_2(x) = C\,e^{(\beta - ik)x} + D\,e^{-(\beta + ik)x}. \tag{19}$$

(Check: $e^{ikx}u_2(x) = Ce^{\beta x} + De^{-\beta x}$, recovering $\psi_{\mathrm{II}}(x)$ of Eq. (14).)

**Boundary conditions on $u_1, u_2$.** Because $u_k(x)$ has the full periodicity of the lattice ($u_k(x+c) = u_k(x)$, a stronger, more convenient statement than the Bloch phase condition on $\psi$ itself), continuity of $u$ and $u'$ can be imposed *directly*, without any extra phase factor, at $x = 0$ and at $x = a \equiv -b$ (identifying the right edge of region 1 with the left edge of region 2 of the *next* cell, which by periodicity of $u$ is the same as the left edge of region 2 of *this* cell):

$$\begin{aligned} &(1) \quad u_1(0) = u_2(0), \\ &(2) \quad u_1'(0) = u_2'(0), \\ &(3) \quad u_1(a) = u_2(-b), \\ &(4) \quad u_1'(a) = u_2'(-b). \end{aligned}$$

These four conditions on the four constants $A, B, C, D$ again require a $4\times4$ determinant to vanish; working it through (exactly as in Section 3 above, now with the Bloch phase already absorbed into $u$ rather than appearing explicitly) yields

$$\boxed{\frac{\beta^2 - \alpha^2}{2\alpha\beta}\sin(\alpha a)\sinh(\beta b) + \cos(\alpha a)\cosh(\beta b) = \cos\big[k(a+b)\big].} \tag{20}$$

With $c \equiv a + b$ this is identical to Eq. (26) derived by the direct-matching route — a useful consistency check, and confirmation that both ways of encoding Bloch's theorem (explicit phase factor on $\psi$, or periodicity built into $u$) must agree.

### 3.4 Applying Bloch's theorem and matching at the boundaries (direct route, recap)

By Bloch's theorem, the wavefunction one period to the left, $\psi(x-c)$ for $0 < x < a$ falling inside the previous barrier region, must equal $\psi_{\mathrm{II}}(x)$ evaluated with $x \to x - c$, up to the Bloch phase:

$$\psi(x) \text{ for } -b < x < 0 \text{ is } \psi_{\mathrm{II}}(x); \qquad \psi_{\mathrm{II}}(x-c) = e^{-ikc}\psi_{\mathrm{II}}(x)\Big|_{\text{shifted by one period back}}. \tag{21}$$

More precisely, continuity of $\psi$ and $\psi'$ is imposed at $x = 0$, and again at $x = a$ where region I must join smoothly onto the *next* barrier region (i.e. region II shifted by $c$, and multiplied by the Bloch factor $e^{ikc}$):

$$\text{(i)}\ \psi_{\mathrm{I}}(0) = \psi_{\mathrm{II}}(0) \qquad A + B = C + D \tag{22}$$

$$\text{(ii)}\ \psi_{\mathrm{I}}'(0) = \psi_{\mathrm{II}}'(0) \qquad i\alpha(A - B) = \beta(C - D) \tag{23}$$

$$\text{(iii)}\ \psi_{\mathrm{I}}(a) = e^{ikc}\,\psi_{\mathrm{II}}(-b) \qquad Ae^{i\alpha a} + Be^{-i\alpha a} = e^{ikc}\left(Ce^{-\beta b} + De^{\beta b}\right) \tag{24}$$

$$\text{(iv)}\ \psi_{\mathrm{I}}'(a) = e^{ikc}\,\psi_{\mathrm{II}}'(-b) \qquad i\alpha\left(Ae^{i\alpha a} - Be^{-i\alpha a}\right) = e^{ikc}\beta\left(Ce^{-\beta b} - De^{\beta b}\right) \tag{25}$$

Equations (22)–(25) are four homogeneous linear equations in the four unknowns $A, B, C, D$. A non-trivial solution exists only if the determinant of coefficients vanishes. Writing out the $4\times4$ determinant and setting it to zero (a lengthy but completely mechanical piece of algebra, carried out in full below) gives the **Kronig–Penney dispersion relation**.

**Working out the determinant (the algebra usually skipped on the board)**

Write the four equations as $M\mathbf{v} = 0$ with $\mathbf{v} = (A, B, C, D)^T$ and

$$M = \begin{pmatrix} 1 & 1 & -1 & -1 \\ i\alpha & -i\alpha & -\beta & \beta \\ e^{i\alpha a} & e^{-i\alpha a} & -e^{ikc}e^{-\beta b} & -e^{ikc}e^{\beta b} \\ i\alpha e^{i\alpha a} & -i\alpha e^{-i\alpha a} & -e^{ikc}\beta e^{-\beta b} & e^{ikc}\beta e^{\beta b} \end{pmatrix}.$$

Setting $\det M = 0$ and simplifying using $\cos\alpha a = \frac{1}{2}(e^{i\alpha a} + e^{-i\alpha a})$, $\sin\alpha a = \frac{1}{2i}(e^{i\alpha a} - e^{-i\alpha a})$, $\cosh\beta b = \frac{1}{2}(e^{\beta b} + e^{-\beta b})$, $\sinh\beta b = \frac{1}{2}(e^{\beta b} - e^{-\beta b})$, one obtains after considerable but elementary trigonometric reduction:

$$\boxed{\cos(kc) \;=\; \cosh(\beta b)\cos(\alpha a) \;+\; \frac{\beta^2 - \alpha^2}{2\alpha\beta}\,\sinh(\beta b)\sin(\alpha a)} \tag{26}$$

This is the exact Kronig–Penney relation. Because $\cos(kc)$ on the left-hand side must lie between $-1$ and $+1$ for $k$ to be real, only those values of $E$ (which determine $\alpha, \beta$) for which the right-hand side also lies in $[-1, 1]$ are allowed: this is precisely the mechanism by which **bands and gaps** arise.

### 3.5 The Kronig–Penney (delta-function) limit

The physically useful and exactly-solvable limit is $b \to 0$, $V_0 \to \infty$, keeping the barrier "strength"

$$P \equiv \lim_{b\to 0,\, V_0\to\infty} \frac{mV_0 ba}{\hbar^2}$$

finite (each barrier becomes a Dirac delta function $V(x) = \dfrac{\hbar^2 P}{ma}\sum_n \delta(x - na)$). In this limit $\beta b \to 0$ so $\cosh(\beta b) \to 1$, $\sinh(\beta b) \to \beta b$, and $\beta^2 b \to \dfrac{2mV_0 b}{\hbar^2} \to \dfrac{2P}{a}$ (using $\beta^2 \approx 2mV_0/\hbar^2$ when $V_0 \gg E$). Substituting into (26) and using $c \to a$:

$$\boxed{\cos(ka) \;=\; P\,\frac{\sin(\alpha a)}{\alpha a} \;+\; \cos(\alpha a)}, \qquad \alpha = \sqrt{2mE/\hbar^2}. \tag{27}$$

This is the standard compact form quoted in every textbook treatment. For a given $\alpha a$ (i.e. given energy $E$) the right-hand side is a definite number; a real Bloch wavevector $k$ exists *only if* that number lies between $-1$ and 1. Equation (27) is transcendental in $E$ and must be solved graphically or numerically.

### 3.6 Graphical solution

Figure 1 plots $f(\alpha a) \equiv P\dfrac{\sin\alpha a}{\alpha a} + \cos\alpha a$ versus $\alpha a$ for $P = 3\pi/2$ (a representative moderate barrier strength). The horizontal band $-1 \le f \le 1$ (shaded) marks the **allowed energy bands**; wherever $|f(\alpha a)| > 1$, no real $k$ exists and that range of $E$ is a **forbidden gap**.

**Limiting cases (useful checks):**

- $P \to 0$ (no barrier): Eq. (27) reduces to $\cos ka = \cos\alpha a \Rightarrow \alpha = k \Rightarrow E = \hbar^2 k^2/2m$, the free-electron parabola, with *no* gaps — as expected, since there is no scattering potential left.
- $P \to \infty$ (impenetrable barriers): the allowed bands shrink to points at $\sin(\alpha a) = 0 \Rightarrow \alpha a = n\pi \Rightarrow E_n = \dfrac{n^2\pi^2\hbar^2}{2ma^2}$, the discrete levels of an isolated infinite square well of width $a$ — i.e. the atoms become totally decoupled, recovering isolated-well ("isolated atom") levels.

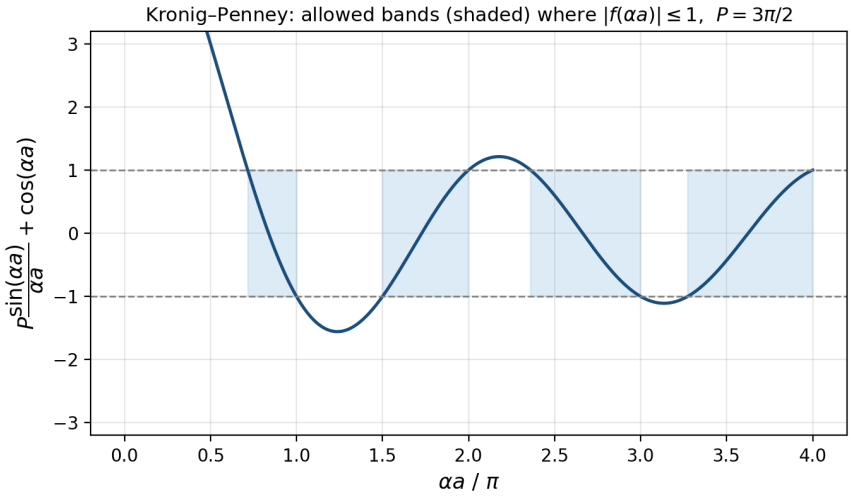



Figure 1: The Kronig–Penney function $f(\alpha a) = P\sin(\alpha a)/(\alpha a) + \cos(\alpha a)$ plotted against $\alpha a/\pi$ for $P = 3\pi/2$. Only the shaded regions ($|f| \le 1$) correspond to allowed energies; the gaps between shaded regions are the forbidden bands. Plotted numerically in Python (matplotlib).

These two limits confirm that the Kronig–Penney model correctly interpolates between the free-electron gas and a set of isolated potential wells, with band gaps appearing for any finite barrier strength $P$.

*[Blank space: insert lecturer's board figure — the piecewise sinusoidal/hyperbolic wavefunction sketch inside a single cell, and/or the specific numerical value of $P$ used in the lecture, if different from the illustrative $P = 3\pi/2$ used above.]*

# 4 The Nearly-Free-Electron (NFE) Model

## 4.1 Setting up the perturbation problem

Here we treat the periodic potential $V(x)$ as a *weak* perturbation on free electrons. Because $V(x)$ has the periodicity of the lattice, it may be expanded as a Fourier series in the reciprocal lattice vectors $G_n = 2\pi n/a$:

$$V(x) = \sum_n V_{G_n}\, e^{iG_n x}, \qquad V_{G_n} = \frac{1}{a}\int_0^a V(x)\, e^{-iG_n x}\, \mathrm{d}x. \tag{28}$$

The unperturbed (free-electron) eigenstates are plane waves $\psi_k^{(0)}(x) = \frac{1}{\sqrt{L}} e^{ikx}$ with $E_k^{(0)} = \hbar^2 k^2/2m$.

## 4.2 Ordinary (non-degenerate) perturbation theory — first pass

Standard first-order perturbation theory gives no energy shift (since $V_{G=0}$ is just a constant reference), and second order gives

$$E_k^{(2)} = \sum_{G\neq 0} \frac{|V_G|^2}{E_k^{(0)} - E_{k-G}^{(0)}}. \tag{29}$$

This expression **diverges** whenever $E_k^{(0)} = E_{k-G}^{(0)}$, i.e. whenever

$$k^2 = (k-G)^2 \implies k = \pm\frac{G}{2} = \pm\frac{n\pi}{a}, \tag{30}$$

which is exactly the Bragg condition for 1-D reflection. This signals a breakdown of naive non-degenerate perturbation theory precisely at the zone boundaries, and tells us we must use **degenerate perturbation theory** there.

### 4.3 Degenerate perturbation theory at the zone boundary

Near $k = \pi/a$ the two free-electron states $|k\rangle$ and $|k-G\rangle$ (with $G = 2\pi/a$, so $k - G = -\pi/a$) are (nearly) degenerate. We diagonalize $H$ in this 2-dimensional subspace. Writing the trial wavefunction as

$$\psi(x) = c_1\, e^{ikx} + c_2\, e^{i(k-G)x},$$

substituting into $H\psi = E\psi$ and projecting onto $\langle k|$ and $\langle k-G|$ gives the secular equations

$$\left(E_k^{(0)} - E\right) c_1 + V_{-G}\, c_2 = 0 \tag{31}$$

$$V_G\, c_1 + \left(E_{k-G}^{(0)} - E\right) c_2 = 0 \tag{32}$$

(using $V_0 \equiv 0$ as the energy zero, and keeping only the dominant Fourier component $V_G$; all other $V_{G'}$ are neglected as they connect states far from degeneracy). Non-trivial $c_1, c_2$ require

$$\det\begin{pmatrix} E_k^{(0)} - E & V_{-G} \\ V_G & E_{k-G}^{(0)} - E \end{pmatrix} = 0 \implies (E_k^{(0)} - E)(E_{k-G}^{(0)} - E) - |V_G|^2 = 0 \tag{33}$$

(using $V_{-G} = V_G^*$ for a real potential, so $V_{-G}V_G = |V_G|^2$). This is a quadratic in $E$:

$$E^2 - \left(E_k^{(0)} + E_{k-G}^{(0)}\right) E + E_k^{(0)} E_{k-G}^{(0)} - |V_G|^2 = 0 \tag{34}$$

$$\implies \quad E_\pm = \frac{E_k^{(0)} + E_{k-G}^{(0)}}{2} \pm \sqrt{\left(\frac{E_k^{(0)} - E_{k-G}^{(0)}}{2}\right)^2 + |V_G|^2}. \tag{35}$$

### 4.4 The energy gap

Exactly at the zone boundary, $k = \pi/a = G/2$, we have $E_k^{(0)} = E_{k-G}^{(0)} \equiv E_0$ (perfect degeneracy), so the square root in (35) collapses to $|V_G|$ and

$$\boxed{E_\pm(k = \pi/a) = E_0 \pm |V_G| \quad \implies \quad E_g \equiv E_+ - E_- = 2|V_G|.} \tag{36}$$

**This is the central result of the nearly-free-electron model:** a gap of magnitude $2|V_G|$ (twice the relevant Fourier component of the lattice potential) opens up at every zone boundary $k = n\pi/a$. Away from the zone boundary, expanding (35) for $|E_k^{(0)} - E_{k-G}^{(0)}| \gg |V_G|$ recovers the unperturbed free-electron result $E_+ \to E_k^{(0)}$, $E_- \to E_{k-G}^{(0)}$, i.e. the bands smoothly rejoin the free-electron parabola far from the zone edge.

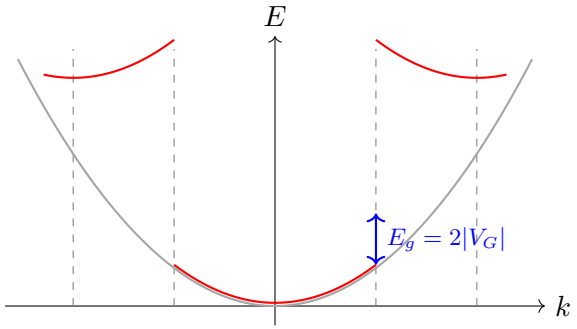
**Also worth deriving: the wavefunctions at the zone edge.** From (31) at $E = E_0 + V_G$ (taking $V_G$ real for simplicity), one finds $c_1 = c_2$, giving

$$\psi_+(x) \propto \cos(\pi x/a),$$

which piles up charge density $|\psi_+|^2 \propto \cos^2(\pi x/a)$ *on* the ion cores (lower energy, attractive potential energy lowers $E$); while at $E = E_0 - V_G$, $c_1 = -c_2$, giving

$$\psi_-(x) \propto \sin(\pi x/a),$$

with charge density concentrated *between* the ions (higher energy). This is the physical origin of the gap: the two standing waves that can be built from $e^{\pm i\pi x/a}$ sample the periodic potential differently, and therefore acquire different potential energy.




Nearly-free-electron bands (red) vs. free-electron parabola (grey); gaps open at $k = \pm n\pi/a$

A numerically accurate version of the same picture (extended parabola on the left, folded reduced-zone bands with gaps on the right) is shown in Figure 2.

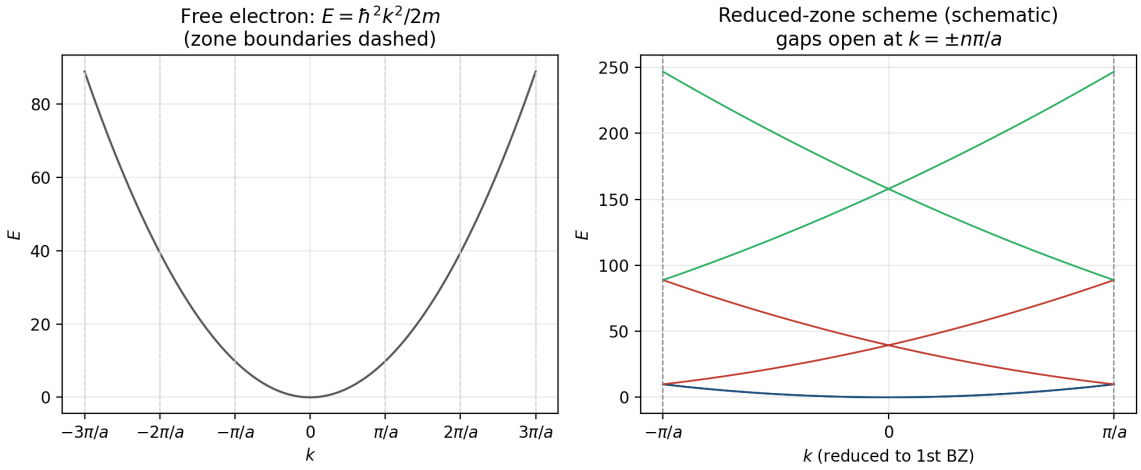



Figure 2: *Left:* the free-electron dispersion $E = \hbar^2 k^2/2m$ with Brillouin-zone boundaries marked. *Right:* schematic of bands folded into the first Brillouin zone; in a real nearly-free-electron calculation each crossing at $k = n\pi/a$ opens into a gap of size $2|V_{G_n}|$ as derived in Eq. (36) (the smooth crossing shown here is the zero-potential limit; superpose the avoided-crossing behaviour by hand if required for the lecture's exact sketch).

# 5 The Tight-Binding (LCAO) Model

## 5.1 Physical picture

The opposite starting point: assume the atoms are far enough apart that each electron is well described by an atomic orbital $\phi(x - x_n)$ centred on lattice site $x_n = na$, with only a small overlap correction from neighbouring atoms (linear combination of atomic orbitals, LCAO).

### 5.2 Bloch sum ansatz

Construct a trial wavefunction that automatically satisfies Bloch's theorem by taking a phased sum over all $N$ atomic sites:

$$\psi_k(x) = \frac{1}{\sqrt{N}} \sum_{n=1}^{N} e^{ikx_n}\, \phi(x - x_n). \tag{37}$$

*Check that this satisfies Bloch's theorem:* shifting $x \to x + a$ and relabelling $n \to n - 1$ in the sum reproduces the same sum multiplied by $e^{ika}$, i.e. $\psi_k(x+a) = e^{ika}\psi_k(x)$, exactly the form (7) required. ✓

### 5.3 Energy expectation value

The Hamiltonian is $H = H_{\text{at}} + \Delta V(x)$, where $H_{\text{at}}$ is the single-atom Hamiltonian (so $H_{\text{at}}\phi(x - x_n) = E_0\phi(x - x_n)$) and $\Delta V$ is the correction due to the presence of the other ion cores. The energy is

$$E(k) = \frac{\langle\psi_k|H|\psi_k\rangle}{\langle\psi_k|\psi_k\rangle}. \tag{38}$$

Assume the atomic orbitals on different sites are orthogonal (or orthogonalized, e.g. Wannier-like), $\langle\phi_n|\phi_m\rangle = \delta_{nm}$, so the denominator is 1. Expanding the numerator,

$$\begin{aligned}\langle\psi_k|H|\psi_k\rangle &= \frac{1}{N}\sum_{n,m} e^{ik(x_n - x_m)} \int \phi^*(x - x_m)\, H\, \phi(x - x_n)\, \mathrm{d}x \\ &= \frac{1}{N}\sum_{n,m} e^{ik(x_n - x_m)}\, H_{mn}.\end{aligned} \tag{39}$$

Define the standard tight-binding matrix elements:

$$-\alpha_{TB} \equiv \int \phi^*(x - x_n)\, H\, \phi(x - x_n)\, \mathrm{d}x \quad \text{(on-site energy correction, same for every } n) \tag{40}$$

$$-\beta_{TB} \equiv \int \phi^*(x - x_{n\pm1})\, H\, \phi(x - x_n)\, \mathrm{d}x \quad \text{(nearest-neighbour "hopping" / overlap integral)} \tag{41}$$

and neglect matrix elements beyond nearest neighbours (they are exponentially small since atomic wavefunctions decay away from the nucleus). Then, because each of the $N$ terms with $m = n$ contributes $-\alpha_{TB}$, and each site $n$ has two nearest neighbours ($m = n \pm 1$) contributing $-\beta_{TB}e^{\mp ika}$:

$$\begin{aligned}\langle\psi_k|H|\psi_k\rangle &= \frac{1}{N}\sum_{n} \Big[(-\alpha_{TB}) + (-\beta_{TB})e^{ik(x_n - x_{n+1})} + (-\beta_{TB})e^{ik(x_n - x_{n-1})}\Big] \\ &= \frac{1}{N}\sum_{n} \Big[E_0 - \alpha_{TB} - \beta_{TB}\left(e^{-ika} + e^{ika}\right)\Big] \\ &= E_0 - \alpha_{TB} - 2\beta_{TB}\cos(ka).\end{aligned} \tag{42}$$

Hence the tight-binding dispersion relation for a 1-D monatomic chain is

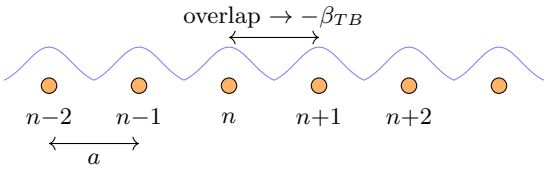
$$\boxed{E(k) = E_0 - \alpha_{TB} - 2\beta_{TB}\cos(ka)} \tag{43}$$

### 5.4 Discussion of the tight-binding band

- $E(k)$ is periodic in $k$ with period $2\pi/a$, as Bloch's theorem requires, and depends on $k$ only through $\cos(ka)$ (even function, as it must be by time-reversal symmetry / inversion symmetry of a monatomic chain).
- The **bandwidth** (max − min of $E(k)$ over the Brillouin zone) is $4\beta_{TB}$: at $k = 0$, $E = E_0 - \alpha_{TB} - 2\beta_{TB}$ (band bottom, for $\beta_{TB} > 0$); at $k = \pm\pi/a$, $E = E_0 - \alpha_{TB} + 2\beta_{TB}$ (band top).
- As the overlap integral $\beta_{TB} \to 0$ (atoms far apart), the bandwidth $\to 0$ and we recover the discrete, dispersionless atomic level $E_0 - \alpha_{TB}$ — consistent with the $P \to \infty$ limit of the Kronig–Penney model found earlier.
- Expanding (43) for small $k$ (band bottom) gives a parabolic approximation

  $$E(k) \approx (E_0 - \alpha_{TB} - 2\beta_{TB}) + \beta_{TB} a^2 k^2,$$

  which defines an **effective mass** $m^* = \hbar^2/(2\beta_{TB}a^2)$ — narrow bands (small $\beta_{TB}$, weak overlap) correspond to *large* effective mass, and wide bands to small effective mass. This connects directly to the free-electron-like behaviour near band edges used later in transport theory.




## 6 Consequences: Brillouin Zones, Metals, Insulators, Semiconductors

### 6.1 Counting states in a band

For a 1-D chain of $N$ atoms with periodic (Born–von Kármán) boundary conditions, $\psi(x+Na) = \psi(x)$ forces

$$e^{ikNa} = 1 \implies k = \frac{2\pi n}{Na}, \quad n = 0, \pm 1, \pm 2, \ldots$$

so allowed $k$-values are spaced by $\Delta k = 2\pi/(Na)$. The width of the first Brillouin zone is $2\pi/a$, so the total number of allowed $k$ states in one band is

$$\frac{2\pi/a}{2\pi/(Na)} = N,$$

i.e. **exactly $N$ states per band** (or $2N$ counting spin). This single counting fact is what decides whether a solid is a metal or an insulator:

- **Metals.** If each atom contributes an *odd* number of valence electrons per band (e.g. one electron per atom, as in the alkali metals), the band is only **partially filled** (e.g. half filled) ⇒ empty states lie immediately above the Fermi level ⇒ an infinitesimal applied field can promote electrons into those empty states, giving high electrical conductivity.
- **Insulators.** If each atom contributes an *even* number of electrons exactly filling the **valence band**, and this is separated from the empty **conduction band** above it by a *large* gap, $E_g \gtrsim$ 3–5 eV, thermal excitation across the gap at ordinary temperatures is completely negligible ($k_B T \approx 0.025$ eV at room temperature) ⇒ effectively no mobile carriers ⇒ non-conducting.

- **Semiconductors.** Same band structure as an insulator (full valence band, empty conduction band) but with a *narrow* gap, $E_g \sim 1$ eV. Thermal energy at room temperature is then sufficient to excite a small but significant number of electrons across the gap, leaving behind **holes** in the valence band; both the excited electrons (in the conduction band) and the holes (in the valence band) contribute to conduction.
- If bands **overlap** in energy (which can happen in 2-D/3-D even when a simple electron count would predict a filled band), electrons spill into the next band before the first is full $\Rightarrow$ also a **metal** (e.g. divalent metals like Mg, Ca).

*[Blank space: insert the lecturer's board diagram comparing filled/empty bands for a metal, insulator and semiconductor, and/or the specific numerical band-gap values quoted in the lecture (e.g. Si, Ge, diamond).]*

### 6.2 Summary table

| Model | Starting point | Key result |
|---|---|---|
| Kronig–Penney | Exactly solvable periodic square/delta wells | $\cos ka = P\sin(\alpha a)/(\alpha a) + \cos \alpha a$; band |
| Nearly-free-electron | Free electrons + weak periodic $V(x)$ | Gap $E_g = 2\lvert V_G\rvert$ opens at each zone bo |
| Tight-binding (LCAO) | Isolated atomic orbitals + weak overlap | $E(k) = E_0 - \alpha_{TB} - 2\beta_{TB}\cos ka$; bandv |

## 7 Consolidated List of Key Results

1. Bloch's theorem: $\psi_k(x) = e^{ikx}u_k(x)$, $u_k(x+a) = u_k(x)$.
2. First Brillouin zone: $k \in (-\pi/a, \pi/a]$; all physics is periodic in $k$ with period $2\pi/a$.
3. Kronig–Penney exact relation: $\cos kc = \cosh(\beta b)\cos(\alpha a) + \dfrac{\beta^2 - \alpha^2}{2\alpha\beta}\sinh(\beta b)\sin(\alpha a)$.
4. Delta-function limit: $\cos ka = P\dfrac{\sin \alpha a}{\alpha a} + \cos \alpha a$.
5. NFE energy gap at zone boundary: $E_g = 2|V_G|$.
6. NFE eigenstates at zone edge: $\psi_+ \propto \cos(\pi x/a)$ (lower energy, charge on ion cores), $\psi_- \propto \sin(\pi x/a)$ (higher energy, charge between ions).
7. Tight-binding dispersion: $E(k) = E_0 - \alpha_{TB} - 2\beta_{TB}\cos ka$; bandwidth $4\beta_{TB}$; effective mass $m^* = \hbar^2/(2\beta_{TB}a^2)$ near band bottom.
8. States per band = $N$ (or $2N$ with spin); electron count per atom (odd/even) plus possible band overlap determines metal vs. insulator vs. semiconductor.

## References (as cited in the course syllabus)


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5. M. S. Vijaya and G. Rangarajan, *Materials Science*, Tata McGraw-Hill, 2004.